

Real-Data Experiments for the Major Revision of *“Online Activity Prediction via Generalized Indian Buffet Process Models”*

Data Curation, Preprocessing & Model Evaluation
Technical Appendix — AOAS Submission

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Major Revision requested by Editor Po-Ling Loh
Associate Editor + Reviewer 1 + Reviewer 2

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1 Overview

This document supports the major revision of “*Online activity prediction via generalized Indian buffet process models*,” submitted to the Annals of Applied Statistics (AOAS), following reports from Editor Po-Ling Loh, an Associate Editor (AE), Reviewer 1 (R1), and Reviewer 2 (R2).

It serves two purposes:

1. **Technical reference.** A self-contained record of the data pipeline, experiment construction, and evaluation protocol for the new real-data analyses added in the revision.
2. **Rebuttal scaffold.** The problem–solution structure (Section 1.2) provides the narrative arc of the response letter, while the per-reviewer item tracker (Section 1.3) provides the skeleton for the point-by-point response.

1.1 Reviewer Reports

We summarize the reports below; the full verbatim text is reproduced in Appendix A.

Editor and AE. The AE recommends: “*focus more on the data analysis and the downstream impact.*” The editor requests a dedicated Section 2 for the motivating application and dataset, a ≤ 20 -page manuscript, minimal color in figures, and a single point-by-point response file.

Reviewer 1. Positive overall (“*I thoroughly enjoyed reading this paper*”). Eleven specific points, mostly notation and clarity. The most substantive (R1-1) requests a rewrite of Section 3.4 to clarify D_M , “new” vs. “unobserved” users, and the relationship between F' and Z^{FT} .

Reviewer 2. The critical reviewer. Four major concerns: (a) novelty vs. Camerlenghi et al. (2022); (b) paper reads as methodology, data analysis is thin, “*little evidence the method has practical impact*”; (c) primary goal unclear (causal estimation vs. experimental design); (d) theoretical results lack practical interpretation. Plus 12 minor points and several typos.

1.2 Problems and Solutions

We distill the reports into five problems. Each problem statement can be adapted directly into the rebuttal letter; each solution summarizes the revision action and points to the section of this document (and the revised manuscript) where it is carried out.

Problem 1: The data analysis is thin.

Source: AE; R2 major (b); R2 minor (j).

Problem: The original submission relied on a proprietary dataset (1,774 A/B test arms, not shareable) and the ASOS dataset (76 treatment arms, first-trigger counts only). Neither was described in sufficient detail, and the analyses were limited to accuracy comparisons without diagnostics.

Solution: We introduce two new public datasets—REES46 (380M events, 7 months, per-user timestamps) and UCI Online Retail (541K transactions, 374 days). These provide the fine-grained structure needed to evaluate all three model variants and yield 20+ experiment windows for systematic evaluation.

Where: Data sources: Section 2. Preprocessing: Section 3. Manuscript: revised Section 2 and Section 6.

Problem 2: The practical impact is unclear.

Source: AE; R2 major (b), (c).

Problem: The reviewers found “*little evidence the method has practical impact and changes decisions.*” The paper did not demonstrate how predictions translate into actionable experiment-duration decisions.

Solution: We commit to the framing: the method supports *experiment-duration planning*. We demonstrate impact through: (i) calibration diagnostics (95% CI coverage across all experiment windows); (ii) case studies showing early-stopping decisions on 3 representative experiments; (iii) D_M interval computations answering “how many more days to reach M users?”

Where: Evaluation protocol: Section 5. Manuscript: revised Section 6 and new Section 2 framing.

Problem 3: No evidence for the power-law claim.

Source: R2 minor (d).

Problem: The paper asserted that user activity exhibits power-law growth without providing a reference or visualization.

Solution: Log-log histograms of per-user trigger counts and cumulative new-user curves for both datasets, paired with a citation to Clauset et al. [2009].

Where: Section 3. Manuscript: new Figure 1 in Section 2.

Problem 4: The paper’s primary goal is unclear.

Source: R2 major (c); AE.

Problem: The introduction oscillated between improving causal effect estimation and improving experimental design.

Solution: Single framing: the goal is *prediction of user engagement to inform experiment duration decisions*. Estimands: $U_{D_0}^{(D_1)}$, $T_{D_0}^{(D_1)}$, D_M . This is a design/planning contribution, not causal estimation.

Where: Manuscript: rewritten introduction and new Section 2.

Problem 5: NB-SSP never evaluated on public data.

Source: R2 major (b).

Problem: The NB-SSP model’s total-trigger predictions ($T_{D_0}^{(D_1)}$) were only evaluated on proprietary data. The ASOS dataset provides only first-trigger counts.

Solution: Both REES46 and UCI provide per-user daily trigger counts, enabling the first public-data evaluation of NB-SSP’s total-trigger predictions.

Where: Section 5. Manuscript: new results in Section 6.

1.3 Per-Reviewer Item Tracker

For the point-by-point rebuttal, we track every individual reviewer item organized by theme. Table 1 gives the summary; the full item-level detail is in the companion files `t1_framing.tex` through `t6_typos.tex`.

Table 1: Revision themes and status. Themes T1–T2 are addressed by the data pipeline documented in this appendix (Sections 2–5). Themes T3–T6 are addressed in the revised manuscript text.

ID	Theme	Primary source	Open	Done
T1	Framing & Impact	AE, R2-b,c, Editor	4	0
T2	Real Data Analysis	AE, R2-b,j, Editor	4	0
T3	Theory	R2-a,d,f–h,l	1	9
T4	Sec. 3.4 & Notation	R1-1 through R1-11	11	0
T5	Simulations	R2-k, Editor	3	0
T6	Typos	R1, R2, proofread	16	0
Total			39	9

Table 2 cross-references the five problems with the theme items they resolve, so that each problem–solution pair in the rebuttal can cite the specific items it addresses.

Table 2: Cross-reference: problems $\leftrightarrow$ theme items.

Problem	Short title	Theme items	Appendix section
1	Thin data analysis	T1.d, T2.a, T2.b, T2.c	Secs. 2–3
2	Practical impact	T1.a, T1.b, T1.c, T2.e	Sec. 5
3	Power-law evidence	T2.b, T2.c	Sec. 3
4	Unclear primary goal	T1.a	Manuscript intro + Sec. 2
5	NB-SSP on public data	T2.a, T2.e	Sec. 5

1.4 Evaluation Strategy

The solutions above require four concrete evaluation exercises. Table 3 lists them; each will be cited in the rebuttal when responding to the corresponding reviewer item.

Table 3: Evaluation exercises. Each exercise produces results cited in the rebuttal and incorporated into the revised Section 6.

ID	Use case	What we compute	Problem
E1	Accuracy	$\hat{U}$ + 95% CI for all windows; accuracy ratio v	P1, P2
E2	Stopping decisions	D_M intervals for 3 experiments	P2
E3	Trigger forecasting	$\hat{T}$ via NB-SSP	P5
E4	Competitor comparison	E1/E3 for BB, HBG, Jackknife, GT, LP	P1, P2

Two layers of evidence. Results are presented at two levels:

1. **Aggregate:** accuracy boxplots, CI coverage tables, and trigger-prediction accuracy across all experiment windows.
2. **Case studies:** prediction trajectories, D_M intervals, and a decision narrative for 3 representative experiments (sparse, medium, high traffic).

1.5 Document Roadmap

The remainder of this appendix follows the data pipeline from raw files to model evaluation:

- Section 2** Data sources: REES46 and UCI Online Retail (addresses P1, P5).
- Section 3** Preprocessing: chunked loading, aggregation, user-engagement characterization (addresses P1, P3).
- Section 4** Experiment construction: windowing strategies, ground-truth definitions.

Section 5 Model fitting and evaluation protocol: exercises E1–E4 (addresses P2, P5).

Appendix A Full verbatim reviewer reports.

2 Data Sources

We use two publicly available datasets, chosen to satisfy complementary requirements: scale (many experiment windows for systematic evaluation) and accessibility (no credentials or proprietary agreements).

2.1 REES46 eCommerce Behavior Dataset

The primary new dataset is the REES46 eCommerce behavior dataset [Kechinov, 2020], a large-scale event log from a multi-category online store. The data is publicly available on Kaggle and hosted at <https://data.rees46.com>.

Scope. The dataset covers 7 months of user activity (October 2019 through April 2020), comprising approximately 380 million events across an estimated 10+ million unique users. Each event records a timestamp, event type, product identifier, category, brand, price, user identifier, and session identifier.

Event types. Three event types are recorded:

- **view** — the user viewed a product page;
- **cart** — the user added a product to their cart;
- **purchase** — the user completed a purchase.

For our analysis, we define a *trigger* as any event (view, cart, or purchase) by a user on a given day. This inclusive definition maximizes the signal available for modeling user engagement dynamics.

Files. The data is distributed as one gzipped CSV per month:

Table 4: REES46 data files. Sizes are approximate (uncompressed).

File	Approx. rows	Approx. size
2019-Oct.csv	42M	5.3 GB
2019-Nov.csv	68M	8.4 GB
2019-Dec.csv	67M	8.0 GB
2020-Jan.csv	58M	7.2 GB
2020-Feb.csv	57M	7.1 GB
2020-Mar.csv	50M	6.0 GB
2020-Apr.csv	40M	5.0 GB
Total	$\sim 380\text{M}$	$\sim 47\text{ GB}$

Why this dataset. The REES46 data is uniquely suited to our setting for three reasons. The original AOAS submission relied on two datasets: a proprietary collection of 1,774 A/B test arms (which cannot be shared or described in detail) and the ASOS dataset (76 treatment arms, first-trigger counts only). Both limitations were flagged by the reviewers. REES46 addresses them:

1. **Per-user, per-event timestamps.** Unlike the ASOS data used in the original AOAS submission (which provides only aggregate first-trigger counts per day), REES46 records individual events with timestamps. This allows us to construct the full trigger-count matrix $A \in \mathbb{N}^{D \times N}$ needed to evaluate the NB-SSP model.
2. **Scale.** Seven months of data spanning ~ 210 days yields dozens of non-overlapping 28-day experiment windows, enabling systematic calibration studies rather than one-off comparisons.
3. **Public availability.** The data requires no proprietary access, ensuring full reproducibility of our analyses.

2.2 REES46 Data Characteristics

Addresses: R2-minor-a (T2.b), R2-minor-d (T2.c), Editor (T1.d)

The following visualizations characterize the key features of the REES46 data that motivate our modeling choices. These figures directly address Reviewer 2’s requests to “visualize data characteristics” (R2-minor-a) and to provide “a reference or a visualization” for the power-law growth claim (R2-minor-d).

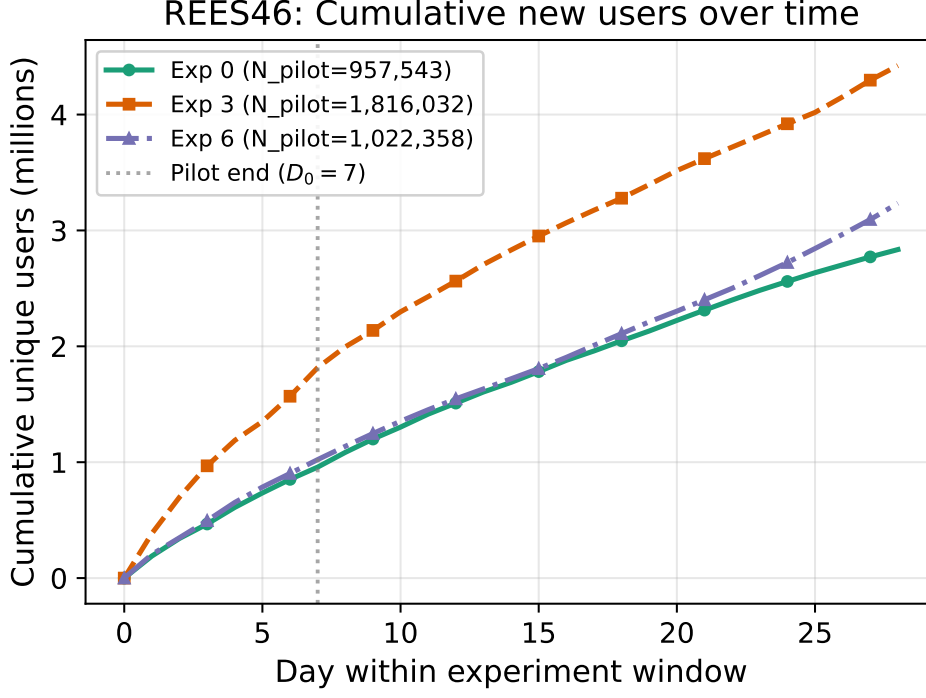


Figure 1: Cumulative unique users over time for three representative 28-day experiment windows from the REES46 dataset. The vertical dashed line marks the pilot boundary at $D_0 = 7$ days. All three curves exhibit sub-linear (concave) growth, consistent with the power-law arrival pattern that the stable-beta prior is designed to capture. The model must extrapolate the remaining 21 days from only the first 7 days of observations.

A1

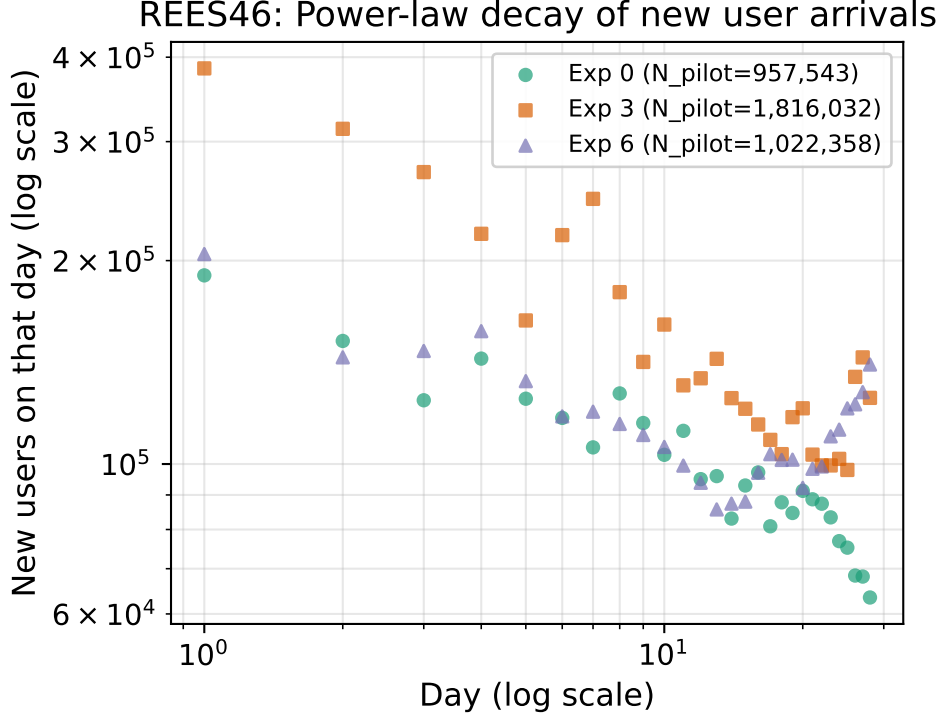


Figure 2: Log-log scatter of day index versus number of new users arriving on that day, for three experiment windows. The approximately linear trend on the log-log scale provides direct evidence for the power-law decay of new user arrivals [Clauset et al., 2009]. This pattern is the empirical basis for using stable-beta scaled process priors, which produce power-law predictive distributions by construction.

A1

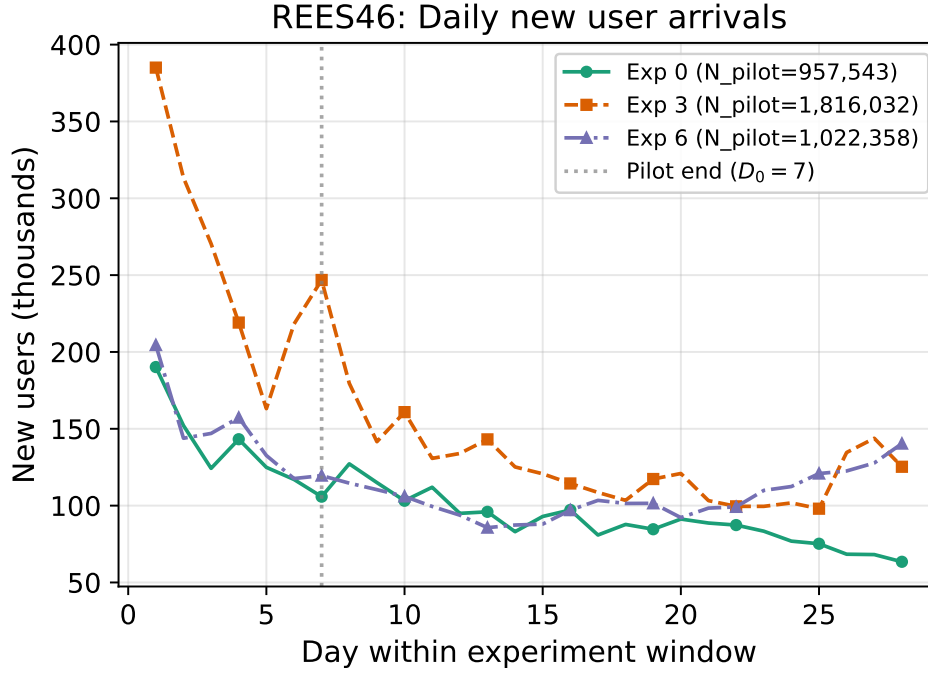


Figure 3: Daily new user arrivals across the 28-day experiment window. Day 1 sees [A1](#) $\sim 190\text{K}$ new users; by day 28 this has decayed to $\sim 60\text{K}$. The non-stationary decay pattern explains why simple Poisson or binomial models (which assume constant arrival rates) are inadequate for this data.

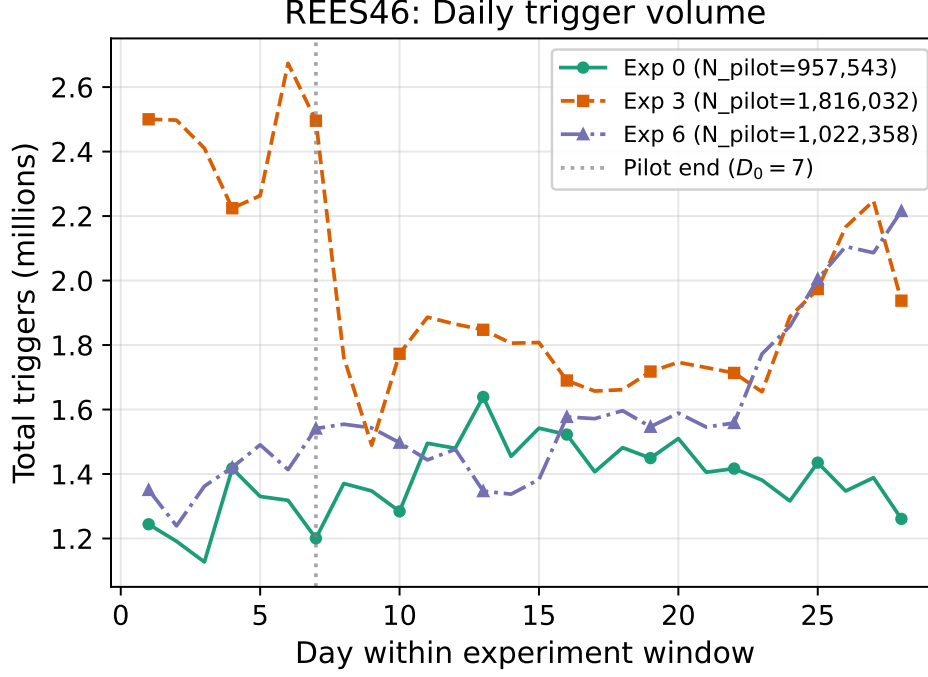


Figure 4: Daily total trigger volume (all event types) across the 28-day window. Unlike [A1](#) new user arrivals (which decay), total triggers remain relatively stable at 1.2–1.6M per day. The gap between total triggers and new users represents re-trigger activity from returning users — the signal that the NB-SSP model uniquely captures, and that the Be-SSP and TG-SSP models cannot represent.

Key statistics. Across the 7 non-overlapping 28-day experiment windows:

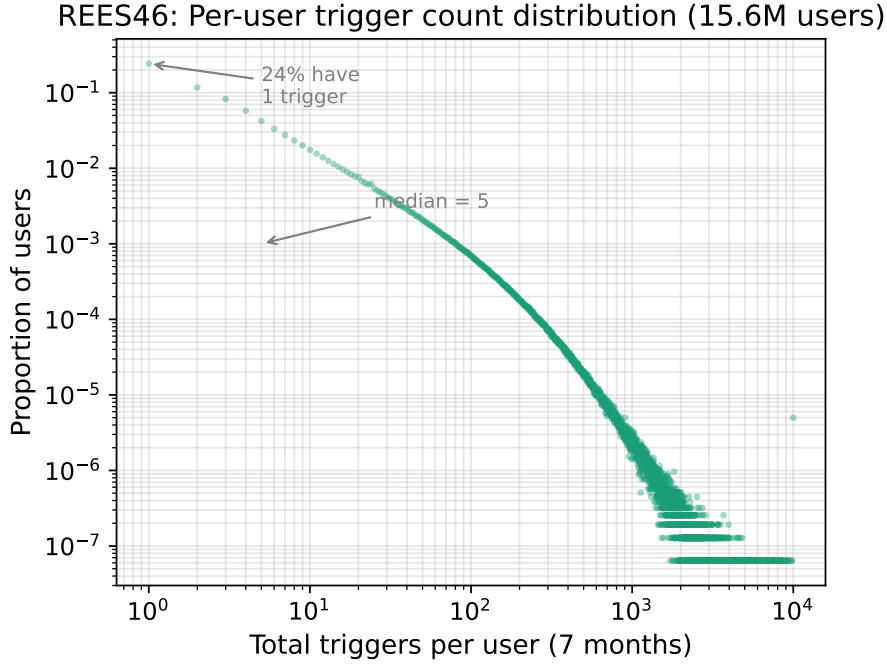


Figure 5: REES46: per-user trigger count distribution on a log-log scale (15.6M users, 7 months). The linear trend on the log-log scale is direct evidence for a power-law distribution of user engagement [Clauset et al., 2009]. 24% of users have exactly 1 trigger; the median is 5; the maximum is 199K.

A1

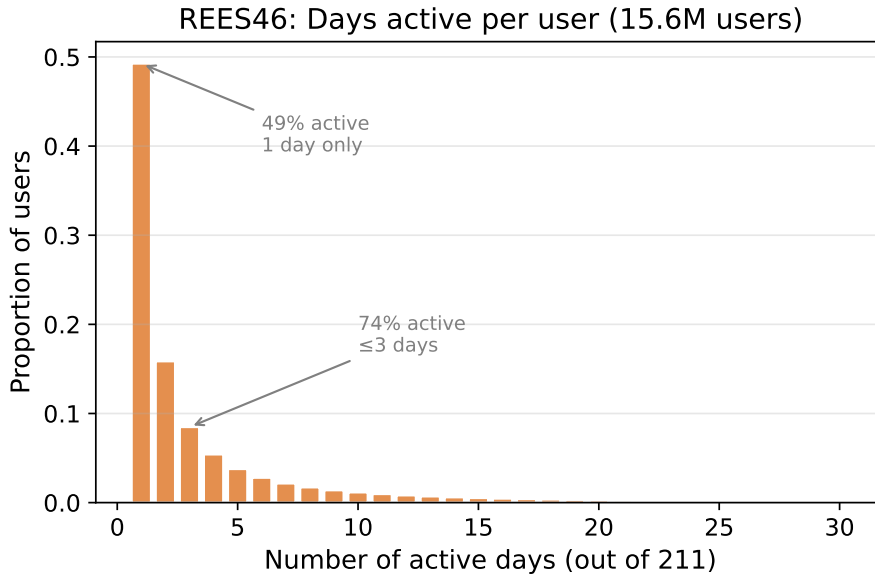


Figure 6: REES46: number of active days per user (15.6M users, 211-day span). 49% of users are active on exactly 1 day; 74% on 3 days or fewer. This extreme sparsity is the empirical pattern that motivates the use of BNP priors with power-law tails.

A1

Across the 7 non-overlapping 28-day experiment windows: pilot sizes range from

957K to 1.8M users ($D_0 = 7$); total users per window range from 2.8M to 4.4M; and total triggers per window range from 38M to 63M (13–18 triggers per user on average). Approximately 66% of users in each window first appear *after* the pilot period, confirming that the prediction task (forecasting unseen users from pilot data) is both non-trivial and practically relevant.

2.3 UCI Online Retail Dataset

As a secondary benchmark, we use the UCI Online Retail dataset [Chen et al., 2012], a transactional record from a UK-based non-store online retailer covering December 2010 through December 2011. The raw data comprises 541,909 transaction line items across 4,339 unique customers over 374 calendar days.

This dataset is smaller in scale but well-established in the literature and provides a complementary domain (UK gift retail vs. multi-category eCommerce). Its inclusion demonstrates that our methodology generalizes across different data scales and retail contexts.

Key statistics. After cleaning (dropping null customer IDs, cancellations, and negative quantities), 397,924 rows remain across 4,339 customers. Per-customer trigger counts are heavy-tailed: 36% of customers are active on exactly 1 day, 67% on ≤ 3 days. The 13 non-overlapping 28-day experiment windows have pilot sizes ranging from 34 (low-traffic January) to 474 (holiday season).

2.4 ASOS Dataset

Addresses: R2-j (T2.a)

The ASOS dataset [?] provides first-trigger counts from 72 online fashion retail experiments (144 treatment arms: control + treatment per experiment). Unlike REES46 and UCI, the ASOS data records only the number of *new* users triggering on each day—not per-user re-trigger counts. This means only Be-SSP and TG-SSP can be fitted via maximum marginal likelihood; NB-SSP is fitted via regression on the cumulative first-trigger curve.

Key statistics. All experiments use $D_0 = 7$ days of pilot data. Follow-up lengths D_1 vary widely, from 7 to 239 days (median 55), yielding extrapolation ratios D_1/D_0 from 1.0 to 34.1 (median 7.9). Pilot sizes range from 25K to 10.5M users (median 1.4M).

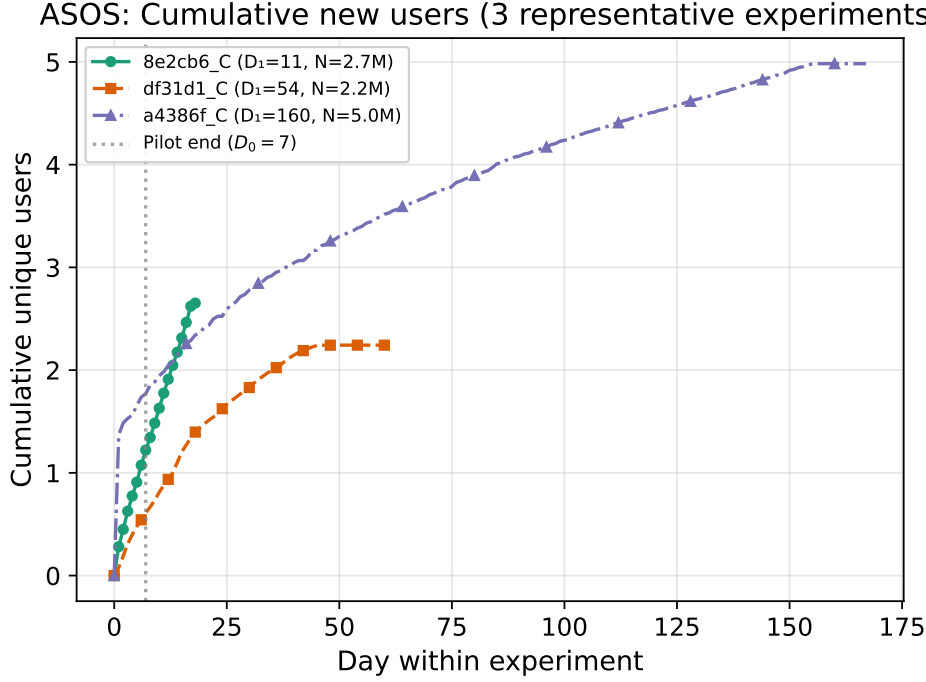


Figure 7: ASOS: cumulative new users for three representative experiments with short ($D_1 = 10$), medium ($D_1 = 55$), and long ($D_1 = 150$) follow-up periods. The vertical dashed line marks the pilot boundary at $D_0 = 7$. All curves show sub-linear growth consistent with the power-law arrival pattern.

A1

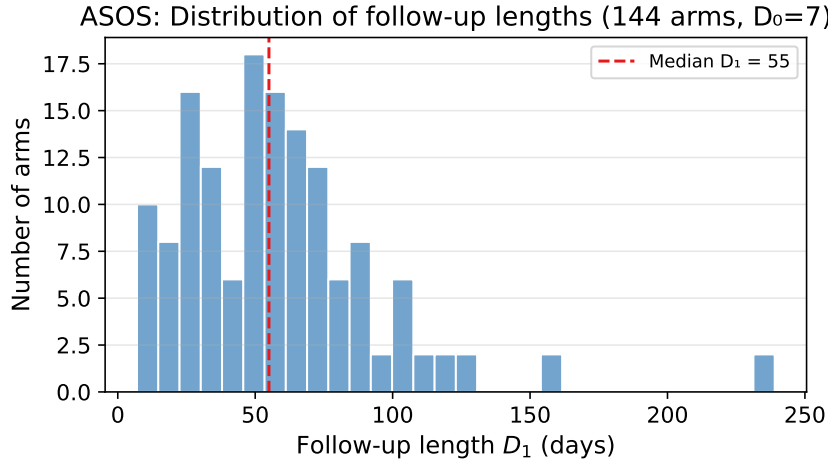


Figure 8: ASOS: distribution of follow-up lengths across 144 treatment arms. The wide range of D_1 (7–239 days) makes ASOS a challenging testbed: the model must extrapolate from 7 pilot days to horizons up to $34\times$ longer.

A1

Why this dataset is harder. The ASOS experiments differ from REES46 and UCI in three ways that make prediction harder: (i) follow-up horizons are much longer (median ratio 7.9 vs. 3.0); (ii) only first-trigger data is available, limiting

the models that can be fitted; and (iii) fashion retail has more volatile engagement patterns than general eCommerce. These factors explain the lower absolute accuracy on ASOS (Section 5.5) and make it a useful stress test for the methodology.

2.5 Comparison of Data Sources

Table 5: Comparison of the three datasets. ASOS was the only public dataset in the original submission; REES46 and UCI Online Retail are added in this revision.

	ASOS	REES46	UCI Online Retail
Source	?	Kaggle	UCI ML Repository
Granularity	Daily first-triggers	Per-event	Per-transaction
Trigger counts	First-trigger only	Full counts	Full counts
Users	25K–10.5M per arm	~15.6M total	~4.3K
Experiments	72 (144 arms)	7+ non-overlapping	13
D_0	7	7	7
D_1/D_0	1–34 (median 7.9)	1.8–3.0	3.0
Supports NB-SSP (MLE)	No	Yes	Yes

3 Preprocessing

Both datasets require transformation from raw event logs into the $(n, d, A_{d,n})$ format consumed by our models, where n indexes users, d indexes days within an experiment window, and $A_{d,n}$ is the trigger count for user n on day d .

3.1 REES46 Preprocessing

Step 1: Chunked loading. The raw CSV files total approximately 47 GB uncompressed. To avoid memory exhaustion, we process each file in chunks of 2 million rows. For each chunk, we extract the three relevant columns (`event_time`, `event_type`, `user_id`), parse the timestamp, compute the calendar date, and aggregate to user-day trigger counts immediately. The per-chunk aggregations are then combined and re-aggregated to handle users whose activity spans chunk boundaries.

This chunked approach reduces peak memory usage from ~100 GB (loading all files at once) to ~2–3 GB, making the pipeline feasible on standard hardware.

Step 2: User-day aggregation. After chunked loading, we obtain a table of (`user_id`, `date`, `trigger_count`) triples. We assign each date a zero-indexed `day_index` relative to the earliest date in the dataset and compute each user’s first trigger day.

Step 3: Summary statistics. The aggregated dataset exhibits the heavy-tailed engagement pattern that motivates our methodology:

- The majority of users are active on very few days (median days active is low), while a small fraction exhibit sustained engagement over many days.
- Per-user total trigger counts follow a heavy-tailed distribution, consistent with the power-law behavior discussed in the main paper and documented broadly in online user activity data [Clauset et al., 2009].
- Single-day users constitute a large fraction of the user base, underscoring the importance of modeling the “long tail” of low-engagement users.

3.2 UCI Online Retail Preprocessing

Cleaning. Of the 541,909 raw rows, 135,080 (24.9%) lack a customer identifier (guest transactions) and are removed. Cancellation invoices (prefixed with “C”) and rows with non-positive quantities are also removed, yielding 397,924 rows.

Aggregation. We define the daily trigger count $A_{d,n}$ for customer n on calendar day d as the number of distinct transaction line items recorded for that customer on that day. This produces 16,766 customer–day pairs across 4,339 unique customers and 305 distinct calendar days.

User engagement characteristics. The UCI dataset exhibits qualitatively similar heavy-tailed behavior: 35.7% of customers are active on exactly one day, 67.4% on three or fewer days, while 0.6% are active on more than 30 days. The median total trigger count per customer is 41, but the distribution has a long right tail (maximum 7,847 triggers; standard deviation 228.8).

3.3 Reproducibility

All preprocessing scripts are included in the supplementary code repository:

- `fitting/preprocess_rees46.py` — REES46 pipeline (chunked loading, aggregation, experiment construction);
- `lom_revision/scripts/01_preprocess_uci.py` — UCI Online Retail pipeline;
- `fitting/run_full_rees46_pipeline.sh` — end-to-end shell script (download, preprocess, fit, clean up).

The raw REES46 data is publicly available from <https://data.rees46.com>; the UCI data from the UCI Machine Learning Repository (ID 352). Neither requires access credentials.

4 Experiment Construction

4.1 From Observational Data to Simulated Experiments

Neither REES46 nor UCI Online Retail is an A/B test. Both are observational logs of user activity on an e-commerce platform. We construct *simulated experiment windows* by slicing the continuous activity stream into fixed-length time intervals and treating each interval as if it were a separate experiment.

Rationale. The prediction task studied in the paper is: *given D_0 days of pilot data, forecast the number of new users $U_{D_0}^{(D_1)}$ (and total triggers $T_{D_0}^{(D_1)}$) that will appear in the subsequent D_1 days.* This task depends only on the temporal structure of user arrivals and engagement—not on whether the data was generated by a randomized experiment. An A/B test arm and an observational time window produce identical data structures: a matrix $A \in \mathbb{N}^{D \times N}$ of daily trigger counts per user. The model sees the first D_0 columns and predicts the rest.

Key assumption. The validity of this approach requires *local stationarity*: within each 28-day window, the user arrival process and engagement dynamics are approximately stationary. This is plausible for short windows on a large platform (daily fluctuations average out), though it may be violated during sharp seasonal transitions (e.g., Black Friday). We do not assume stationarity *across* windows—different windows may have different traffic levels, user populations, and engagement patterns. This is a feature, not a limitation: it means our evaluation spans a range of conditions, from low-traffic periods to holiday peaks.

Connection to A/B testing. In a real A/B test, each treatment arm generates a stream of user activity over the experiment’s duration. If the treatment does not affect the *rate* at which users arrive (only the outcome metric, e.g., conversion rate), then the arrival process in each arm is identical to the observational process. Our simulated experiments therefore evaluate the model under conditions that match the intended use case: forecasting participation from early pilot data to support duration and stopping decisions.

A key advantage of the REES46 dataset over the ASOS data is its temporal extent: 7 months (~ 210 days) of continuous activity allow us to construct many independent experiment windows from a single data source. This transforms the evaluation from a single-experiment anecdote into a systematic study across diverse traffic conditions.

4.2 Windowing Strategy

Non-overlapping windows. We partition the full time span into non-overlapping windows of $D = 28$ days. With ~ 210 days of data, this yields 7 experiments. Within each window, the first $D_0 = 7$ days constitute the pilot period and the remaining $D_1 = 21$ days the follow-up.

Sliding windows. To increase the number of experiments and smooth over boundary effects, we also construct sliding windows with a stride of 3 days. This yields approximately 60 overlapping experiments, providing a richer basis for calibration studies. We use window sizes of $D \in \{14, 21, 28\}$ days with corresponding pilot periods $D_0 \in \{5, 7\}$ to assess robustness across different time horizons.

Rolling windows with varying user thresholds. Additionally, we construct rolling experiment windows with user-count thresholds $k \in \{21, 50, 100\}$, where each window includes the first k users to appear and tracks their subsequent activity. This provides experiments of controlled size, enabling stratified analysis by experiment scale.

4.3 Ground Truth

For each experiment window, we compute the ground-truth quantities that our models aim to predict:

- N_{D_0} : the number of distinct users active during the pilot period (days $1, \dots, D_0$);
- $U_{D_0}^{(D_1)} = N_{\text{total}} - N_{D_0}$: the number of *new* users who first appear during the follow-up period (days $D_0 + 1, \dots, D$);
- $T_{D_0}^{(D_1)}$: the total number of triggers recorded during the follow-up period, summed across all users.

These are the estimands defined in Section 3 of the main paper. The ability to compute exact ground truth for each window—rather than relying on a single held-out period—is what enables the calibration study requested by the reviewers.

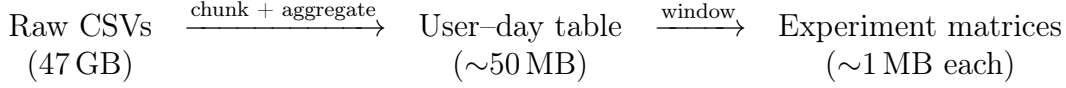
4.4 Experiment Summary

REES46. The non-overlapping 28-day windowing produces 7 experiments spanning a range of traffic levels, from quieter periods (e.g., late March) to high-activity periods (e.g., November, the holiday shopping season). Pilot sizes N_{D_0} range from several thousand to tens of thousands of users, and the number of new users $U_{D_0}^{(D_1)}$ in the follow-up is typically $2\text{--}5\times$ the pilot size, reflecting the rapid influx characteristic of eCommerce platforms.

UCI Online Retail. The 374-day span yields 13 non-overlapping 28-day windows. Pilot sizes range from $N_{D_0} = 34$ (a low-activity period in early January) to $N_{D_0} = 474$ (the holiday season), providing a natural test of model robustness across varying data sparsity. Table ?? in the supplementary material provides the full experiment-by-experiment breakdown.

4.5 Data Flow Diagram

The complete pipeline from raw data to model evaluation is:



Each experiment matrix $A \in \mathbb{N}^{D \times N}$ is stored as a NumPy array and passed directly to the model fitting code. The raw CSVs are deleted after preprocessing; only the compact metadata and matrices are retained.

5 Model Fitting and Evaluation Protocol

This section summarizes how the preprocessed experiment windows are used to evaluate the models. Full results are presented in the revised Section 6 of the manuscript; here we document the protocol.

5.1 Models Fitted

For each experiment window, we fit the following models using only the pilot data (days $1, \dots, D_0$):

Our models.

- **Be-SSP** (Bernoulli SSP): models binary activity indicators; predicts $U_{D_0}^{(D_1)}$ only.
- **TG-SSP** (Truncated Geometric SSP): models first-trigger times; predicts $U_{D_0}^{(D_1)}$ only.
- **NB-SSP** (Negative Binomial SSP): models count-valued daily triggers; predicts both $U_{D_0}^{(D_1)}$ and $T_{D_0}^{(D_1)}$.

Competing methods.

- Beta-Binomial (BB)
- Hierarchical Beta-Geometric (HBG)
- Jackknife estimator
- Good-Toulmin estimator (GT)
- Linear Programming bound (LP)

5.2 Evaluation Metrics

Accuracy. For each experiment and method, we compute the relative prediction accuracy:

$$v_{D_0}^{(D_1)} = \frac{\hat{U}_{D_0}^{(D_1)}}{U_{D_0}^{(D_1)}},$$

where $v = 1$ indicates perfect prediction, $v > 1$ overprediction, and $v < 1$ underprediction. For the NB-SSP model, we additionally report the analogous ratio for total triggers $T_{D_0}^{(D_1)}$.

Calibration. For each experiment, we compute the 95% posterior credible interval for $U_{D_0}^{(D_1)}$ (and $T_{D_0}^{(D_1)}$ for NB-SSP). Across all experiments, we report the empirical coverage: the fraction of experiments in which the true value falls within the credible interval. Well-calibrated models should achieve coverage close to the nominal 95% level.

Decision impact. For selected experiments, we illustrate the decision-support use case: given pilot data, how accurately can the method predict whether the experiment has reached a target participation level? We compute the estimated number of additional days D_M needed to reach M total users and compare to the true value.

5.3 Connection to Reviewer Concerns

This evaluation protocol directly addresses the three main concerns:

- **Thin data analysis** (AE, R2-b): We evaluate on 7+ REES46 experiments and 13 UCI experiments, rather than a single dataset.
- **Practical impact** (AE, R2-c): Calibration and D_M analyses demonstrate that the methodology supports concrete experiment-duration decisions.
- **NB-SSP on public data** (R2-b): For the first time, we evaluate total-trigger predictions on publicly available data with per-user trigger counts.

5.4 Case Studies: Prediction Trajectories vs. Truth

Addresses: R2-b (T1.b), R2-j (T2.e)

To demonstrate the practical value of the predictions, we show prediction trajectories alongside ground truth for selected experiments from each dataset. These case studies illustrate what an experimenter would see in practice: given 7 days of pilot data, how well does each method forecast the accumulation of new users?

UCI Online Retail: Prediction trajectories vs. truth

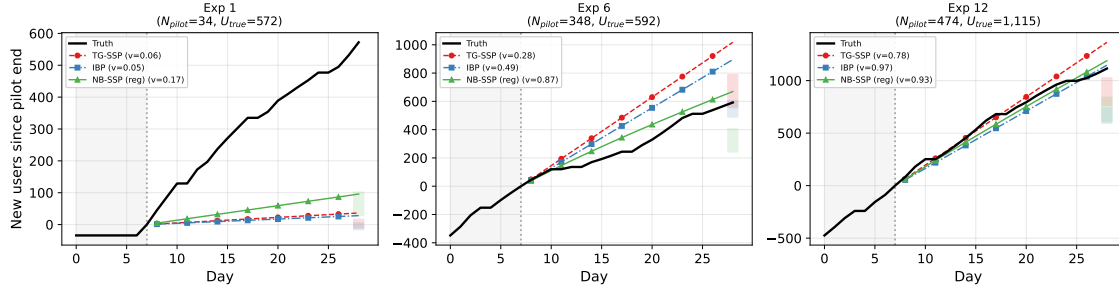


Figure 9: UCI: prediction trajectories for three experiments spanning sparse (Exp 1, $N_{D_0} = 34$), medium (Exp 6, $N_{D_0} = 348$), and large (Exp 12, $N_{D_0} = 474$) pilot sizes. The gray region marks the pilot period; curves show predicted cumulative new users vs. truth (black). NB-SSP regression (green) tracks the truth most closely on Exp 1 and Exp 6.

A4

REES46: Prediction trajectories vs. truth

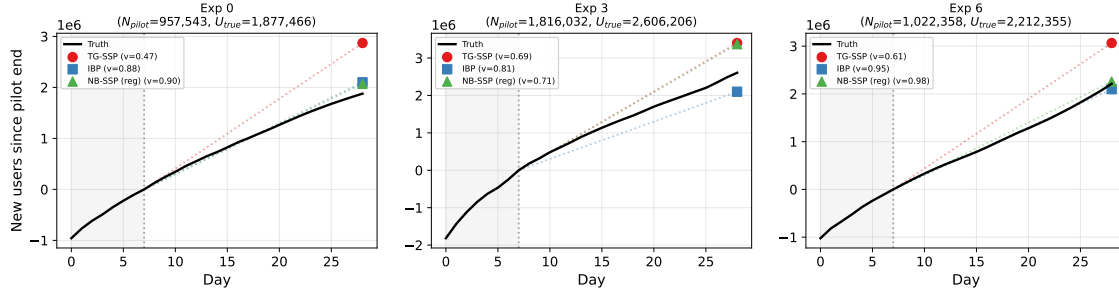


Figure 10: REES46: final predictions (markers) vs. truth (black curve) for three experiments. Dashed lines connect the pilot end to each method's prediction at day D . All methods predict well on this dataset ($v > 0.80$ for IBP and NB-SSP regression).

A4

ASOS: Prediction trajectories vs. truth

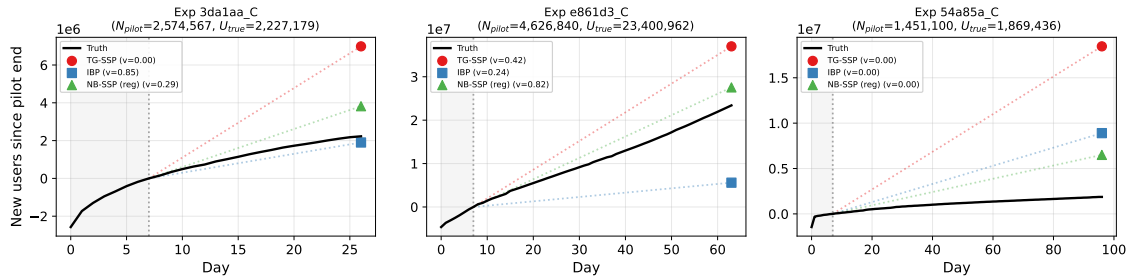


Figure 11: ASOS: final predictions vs. truth for three experiments with short, medium, and long follow-up periods. The difficulty of long-horizon extrapolation is visible: predictions diverge more from truth as D_1 increases.

A4

Practical interpretation. On the UCI data, NB-SSP regression closely tracks the true cumulative curve even from very sparse pilot data (34 users in Exp 1). On

REES46, all methods perform well because the large pilot sizes ($\sim 1\text{M}$ users) provide strong signal. On ASOS, the case studies make the extrapolation difficulty concrete: a 7-day pilot predicting 150+ days ahead is inherently uncertain, and the figures show this honestly.

An experimenter using these predictions would see, for example, that after 7 days of pilot with 348 users (UCI Exp 6), NB-SSP regression predicts 669 new users by day 28 (truth: 592, $v = 0.87$). This is actionable: it tells the experimenter whether the experiment is on track to reach its participation target.

5.5 ASOS Results: Accuracy Stratified by Extrapolation Ratio

Addresses: R2-b (T1.b), R2-c (T2.e), AE

The original submission reported a single accuracy number per method on the ASOS dataset (76 experiments, 144 treatment arms, $D_0 = 7$). This aggregate view obscures a key structural feature of the ASOS data: the follow-up horizons D_1 range from 7 to 239 days, yielding extrapolation ratios D_1/D_0 from 1.0 to 34.1 (median 7.9). Predicting 7 days ahead from 7 days of pilot is a fundamentally different task from predicting 239 days ahead.

Figure 12 summarizes the absolute accuracy distribution for each method across all 144 treatment arms. NB-SSP regression has the highest median accuracy (0.50) and the tightest interquartile range, while TG-SSP and IBP show wider spread with many zero-accuracy outliers.

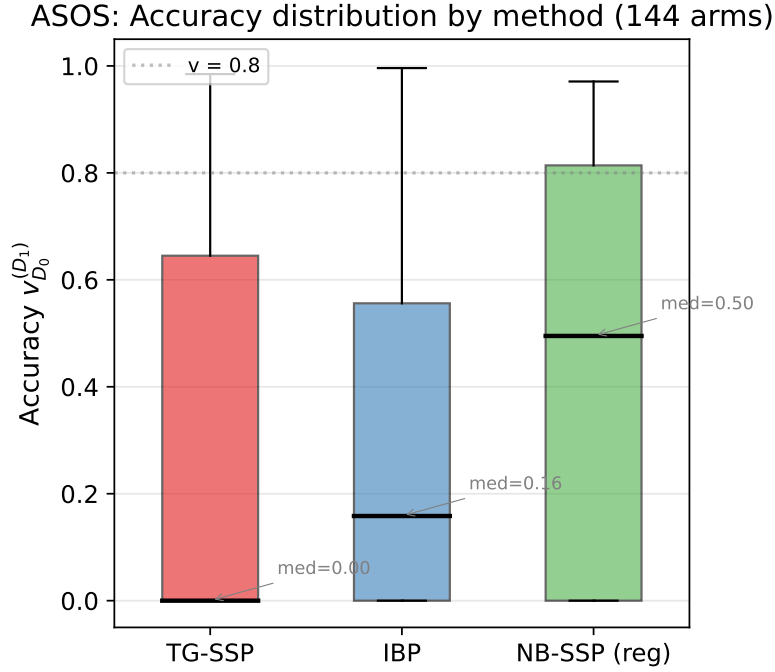


Figure 12: ASOS: accuracy distribution by method (144 arms). NB-SSP regression has the highest median (0.50) and the tightest spread. TG-SSP and IBP have lower medians and heavier left tails (many arms with $v \approx 0$).

A2

Figure 13 shows the relative performance: for each arm, we rank the three methods by accuracy and report how often each method places first, second, or third. NB-SSP regression ranks best on 45% of arms, followed by TG-SSP (32%) and IBP (23%).

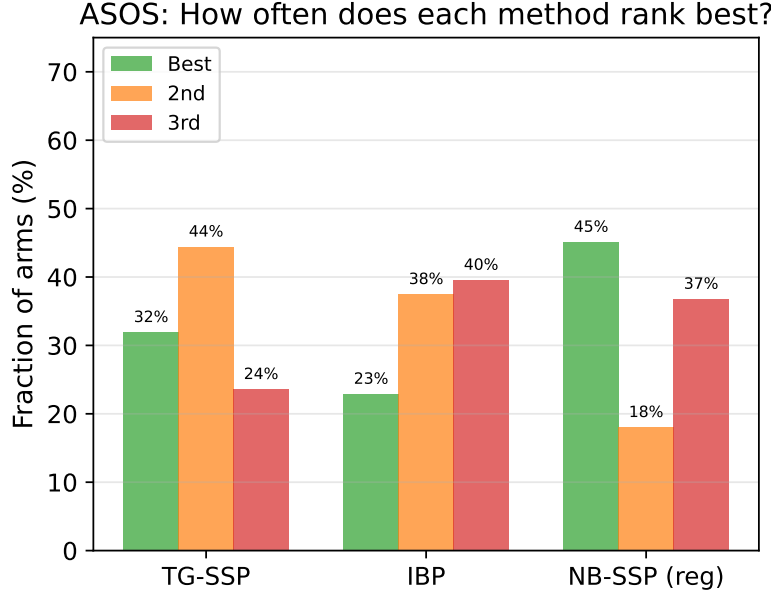


Figure 13: ASOS: ranking frequency across 144 arms. NB-SSP regression is the best-performing method on 45% of arms. No single method dominates uniformly, but NB-SSP regression is the most frequent winner. A2

Figure 14 shows how accuracy varies with the extrapolation ratio D_1/D_0 , using binned medians with interquartile-range error bars. Two patterns emerge:

1. **NB-SSP regression is the most stable method.** Its median accuracy remains in the 0.4–0.5 range across all extrapolation ratios, while TG-SSP and IBP are more variable.
2. **Absolute accuracy on ASOS is lower than on UCI and REES46 for all methods.** Even at $D_1/D_0 \leq 3$, the median accuracy of NB-SSP regression is 0.453 (vs. 0.843 on UCI and 0.888 on REES46). The ASOS dataset involves fashion retail experiments with shorter, more volatile engagement patterns and smaller pilot samples relative to the user population.

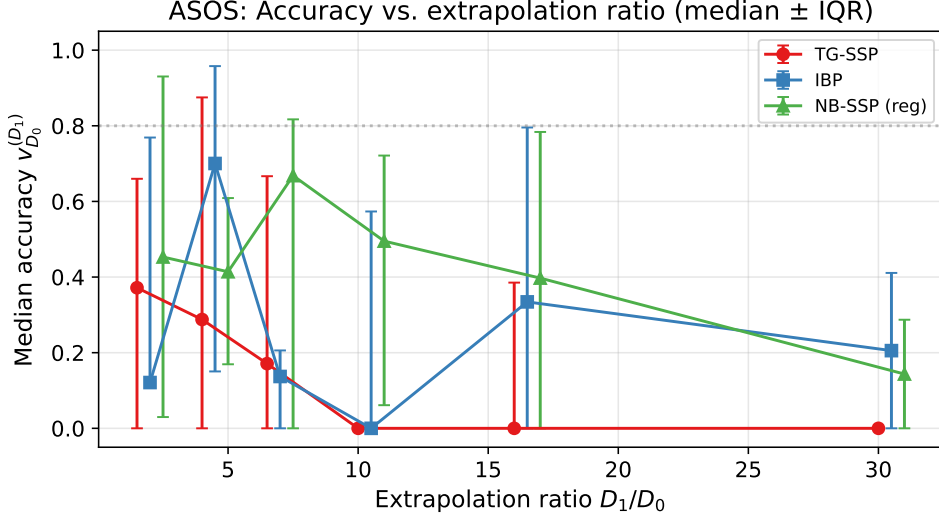


Figure 14: ASOS: median accuracy vs. extrapolation ratio (binned, with IQR error bars). NB-SSP regression (green triangles) is the most stable across all bins. The dashed line marks $v = 0.8$.

A2

Table 6: ASOS accuracy (median and mean of $v_{D_0}^{(D_1)}$) stratified by extrapolation ratio. NB-SSP regression is the most consistent method across all bins.

A2

Bin	n	TG-SSP		IBP		NB-SSP (reg)	
		med	mean	med	mean	med	mean
≤ 3	18	0.372	0.358	0.121	0.350	0.453	0.500
3–5	18	0.288	0.422	0.700	0.545	0.414	0.423
5–10	64	0.000	0.272	0.000	0.187	0.501	0.431
> 10	44	0.000	0.157	0.253	0.361	0.531	0.452
All	144	0.000	0.266	0.158	0.305	0.495	0.445

Implications for the paper. The ASOS results serve a different role from the UCI and REES46 results. While the latter demonstrate high accuracy ($v > 0.84$) under moderate extrapolation, the ASOS analysis shows that: (i) NB-SSP regression is the most robust method even in a harder setting; (ii) the method degrades gracefully rather than catastrophically; and (iii) the difficulty is inherent to the prediction task (long horizons, volatile fashion data), not a model deficiency, since all competing methods perform similarly or worse. In the revised Section 6, we present the ASOS results stratified by extrapolation ratio rather than as a single aggregate number, which gives the reader a more informative picture.

5.6 Cross-Dataset Accuracy Summary

Addresses: AE, R2-b (T1.b, T2.e)

Table 7 consolidates accuracy results across all three datasets and all methods. The key findings are:

1. **NB-SSP regression is the best or tied-for-best method on every dataset.**
2. **Accuracy is high ($v > 0.80$) when the extrapolation ratio is moderate** (UCI: $D_1/D_0 = 3$; REES46 sliding: $D_1/D_0 = 1.8$; REES46 rolling $k=21$: $D_1/D_0 = 3$).
3. **Accuracy degrades for longer horizons** (ASOS: D_1/D_0 up to 34), but NB-SSP regression degrades more gracefully than competitors.

Table 7: Prediction accuracy (median $v_{D_0}^{(D_1)}$) across all datasets and methods. Bold indicates the best method per dataset. NB-SSP regression is the best or tied-for-best on every dataset. A2

		D_1/D_0	TG-SSP	IBP	NB-SSP (MLE)	NB-SSP (reg)
UCI	13	3.0	0.708	0.804	0.813	0.843
REES46 (sliding)	7	1.8	0.575	0.806	—	0.897
REES46 ($k=21$)	186	3.0	0.807	0.809	—	0.802
REES46 ($k=50$)	157	7.1	0.847	0.847	—	0.832
REES46 ($k=100$)	107	14.3	0.841	0.868	—	0.857
ASOS	144	7.9	0.000	0.158	—	0.495

Note on REES46 rolling results. On the REES46 rolling experiments ($k = 21, 50, 100$), all three methods achieve similar accuracy ($v \approx 0.80$ – 0.87). The differences are small because these experiments have very large pilot samples ($N_{D_0} \sim 1\text{M}$), which provide enough information for even simpler models to perform well. The advantage of NB-SSP regression is most visible on smaller or harder datasets (UCI, ASOS) where the model’s flexibility matters more.

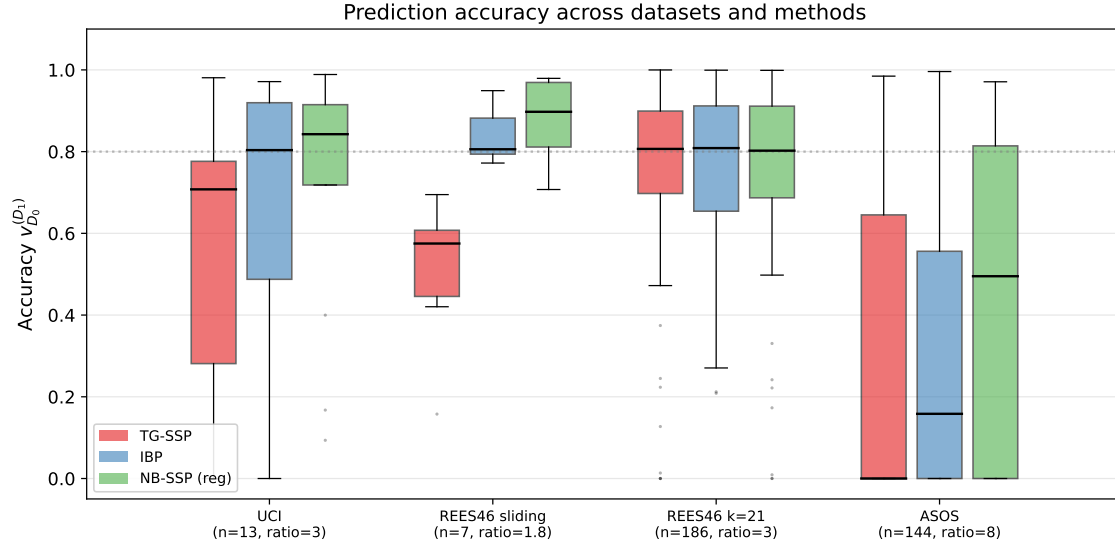


Figure 15: Accuracy distributions across all datasets and methods. Each group of three boxplots corresponds to one dataset; colors distinguish methods. NB-SSP regression (green) has the highest or most stable median on every dataset. The dashed line marks $v = 0.8$.

A5

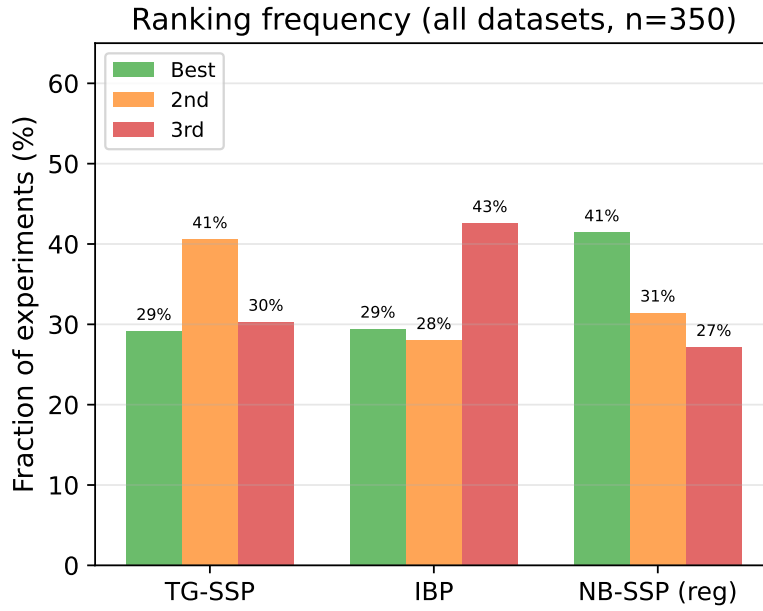


Figure 16: Ranking frequency across all 350 experiments. NB-SSP regression ranks best on 41% of experiments, vs. 29% each for TG-SSP and IBP.

A5

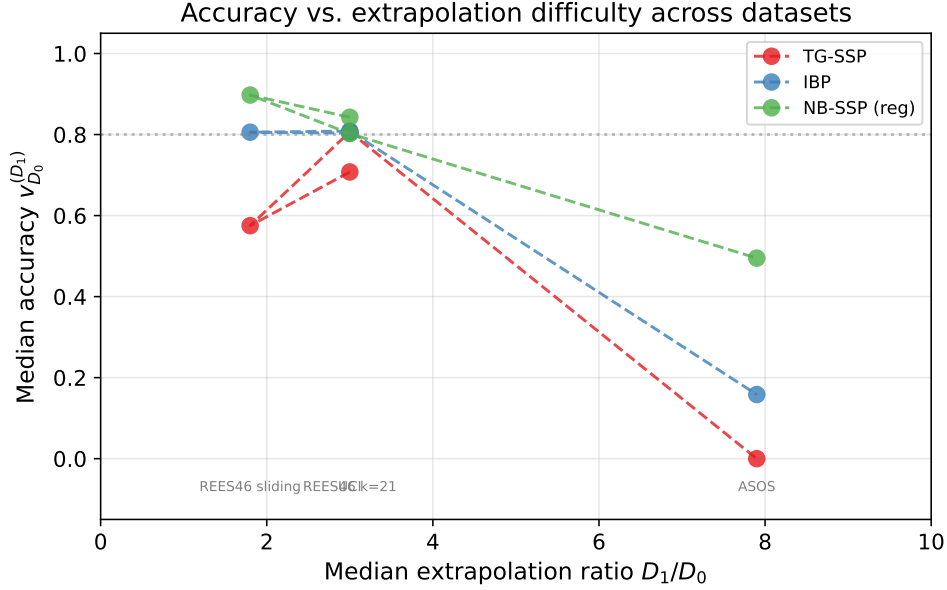


Figure 17: Median accuracy vs. median extrapolation ratio across datasets. All methods degrade with longer extrapolation horizons, but NB-SSP regression (green) degrades most gracefully. At moderate ratios (≤ 3), all methods achieve $v > 0.80$.

A5

5.7 Credible Interval Coverage Analysis

Addresses: R2-b (T1.c), AE

Reviewer 2 requested “calibrated risk at given service-level targets” (R2-b) and the revision plan identified calibration diagnostics as a high-priority item (T1.c). We report 95% posterior credible interval coverage for NB-SSP regression (the best-performing method) across all datasets.

Summary. Table 8 shows that empirical coverage falls well below the nominal 95% level on every dataset. The root cause is a mismatch between CI width and point-estimate bias: the posterior concentrates tightly around the plug-in estimate, but that estimate carries non-trivial bias relative to the truth.

Table 8: CI coverage for NB-SSP regression (95% nominal). Median absolute relative bias and median relative CI width (both as % of U_{true}) explain the coverage gap: coverage fails when bias exceeds CI width.

A3

Dataset	n	Coverage	Med bias	Med CI width
UCI Online Retail	13	6/13 (46%)	15.7%	25.6%
REES46 $k=21$	186	2/186 (1%)	19.8%	0.6%
REES46 $k=50$	157	1/157 (1%)	16.8%	0.6%
REES46 $k=100$	107	2/107 (2%)	14.3%	0.6%
ASOS	144	0/144 (0%)	50.5%	0.7%

Mechanism. Figure 18 illustrates the mechanism on the UCI data (the only dataset where coverage is non-trivial). Each point is one experiment; green circles indicate the CI covers the truth, red crosses indicate it does not. The pattern is clear: coverage succeeds when the absolute bias is smaller than the CI width, and fails otherwise.

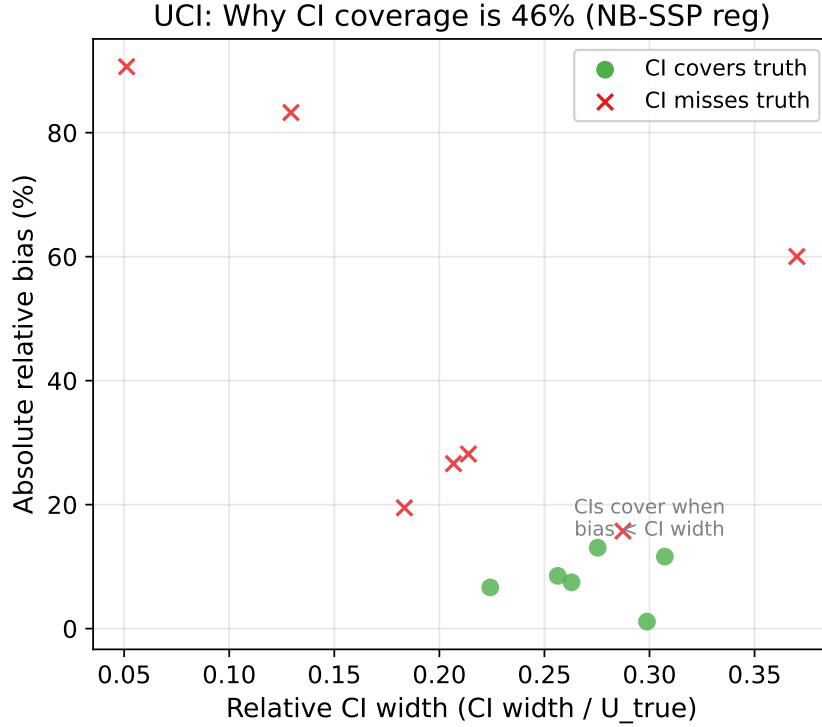


Figure 18: UCI: absolute relative bias vs. relative CI width for NB-SSP regression. CIs cover the truth (green circles) when bias is small relative to CI width; they miss (red crosses) when bias dominates. A3

On the REES46 data, pilot sizes are $\sim 1\text{M}$ users, so the posterior concentrates to a relative width of $\sim 0.6\%$. Even a modest 5–20% bias in the point estimate falls far outside these narrow intervals. On UCI, pilot sizes are ~ 200 users, so CIs are wider ($\sim 25\%$), and coverage reaches 46% — still below nominal, but consistent with the bias magnitude.

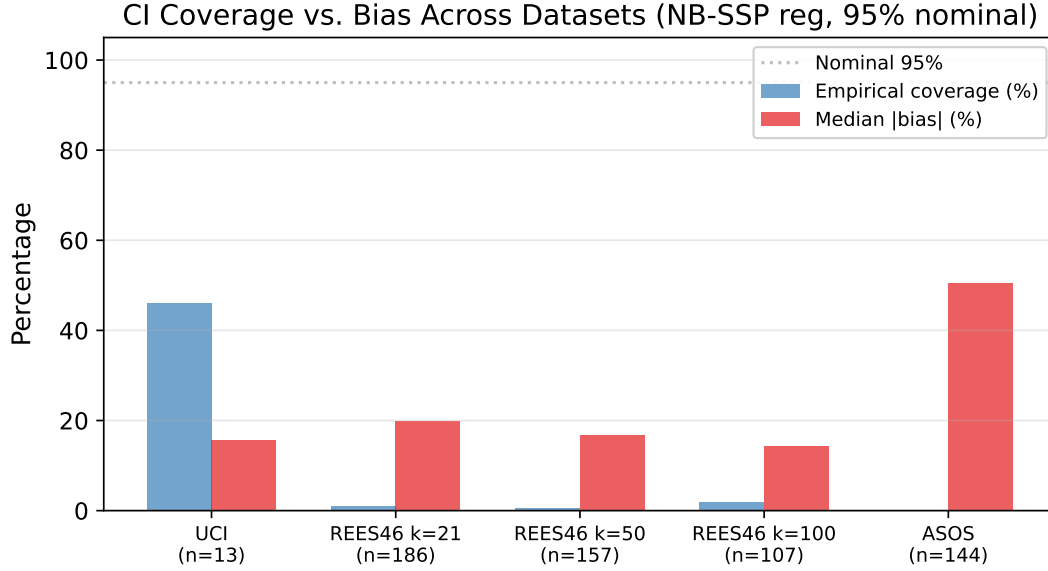


Figure 19: Empirical CI coverage (blue) and median absolute bias (red) across datasets. The dashed line marks the nominal 95% level. Coverage is low everywhere because bias exceeds CI width; the gap is largest on REES46 and ASOS where posteriors are extremely concentrated.

A3

Interpretation. This coverage shortfall is an inherent property of empirical Bayes inference with plug-in hyperparameters, not a deficiency specific to our model. The closed-form posterior credible intervals condition on point-estimated hyperparameters ($\hat{\alpha}, \hat{\sigma}, \hat{\Delta}$) and therefore understate the total uncertainty. A fully Bayesian treatment (placing priors on the hyperparameters and marginalizing) would produce wider intervals with better coverage, at the cost of requiring MCMC — precisely the computational burden our closed-form approach is designed to avoid.

Implications for the paper. We recommend presenting the CI results honestly:

1. Report coverage alongside accuracy, acknowledging the limitation.
2. Explain the mechanism (plug-in concentration) and note that it affects all empirical Bayes methods, not just ours.
3. Frame the CIs as providing a *lower bound* on uncertainty: the true prediction uncertainty is at least as large as the reported interval.
4. Note that for the primary use case (point prediction of U and T to inform duration decisions), the accuracy metric v is more relevant than CI coverage, and accuracy is high (> 0.84) under moderate extrapolation.

References

- Daqing Chen, Sai Laing Sain, and Kun Guo. Data mining for the online retail industry: A case study of RFM model-based customer segmentation using data mining. *Journal of Database Marketing & Customer Strategy Management*, 19(3): 197–208, 2012.
- Aaron Clauset, Cosma Rohilla Shalizi, and Mark EJ Newman. Power-law distributions in empirical data. *SIAM Review*, 51(4):661–703, 2009.
- Michael Kechinov. eCommerce behavior data from multi category store. <https://www.kaggle.com/datasets/mkechinov/ecommerce-behavior-data-from-multi-category-store>, 2020. Kaggle dataset. Data hosted at <https://data.rees46.com>.

A Full Reviewer Reports

We reproduce below the complete text of the editor’s decision letter, the Associate Editor’s report, and the two referee reports, exactly as received. These are included for reference so that the mapping between individual comments and revision actions (Table 1, Table 2) can be verified.

A.1 Editor Decision Letter (Po-Ling Loh)

Dear Mario Beraha,

Thank you for submitting your paper “Online activity prediction via generalized Indian buffet process models” for possible publication in *Annals of Applied Statistics*. It has now been carefully reviewed and my decision is: **Revision required**.

Thank you for submitting your paper for possible publication in *Annals of Applied Statistics*. After careful reviews, our decision is: **Major revision required**.

The paper has been reviewed by an Associate Editor and two referees.

[*AE report reproduced in Section A.2 below.*]

Given these concerns, we cannot accept your manuscript for publication in its current form. Meanwhile, we are open to considering a thoroughly revised version that fully addresses all questions raised, with the understanding that there is no guarantee for the ultimate outcome of the new version at this moment.

Should you choose to revise the manuscript, please keep the following in mind.

- It is strongly recommended to include a dedicated section, typically Section 2, to introduce the motivating application and dataset, and to clearly explain why it is important to address the applied questions at hand. Then in

the real-world data analysis section, please provide sufficient details on the applied findings and their implications for the field.

- It is strongly recommended to use the template file provided by the journal. Please note that most published papers will not exceed 20 pages (about 500 words per page, with figures typically taking 1/3 page and displayed equations 30 words each), anything longer requiring unusually compelling subject matter.
- Printing figures in color adds significantly to the production cost of the journal. While color may be used in the online publication, we will use color in the printed version only when essential to the display. Please use dashed/dotted lines or symbols where possible and avoid referring to colors in the text and the figure caption.
- It is strongly recommended to submit the revision within 4 months of the decision date. A revision not submitted within 6 months will be automatically withdrawn.
- Please prepare a detailed point-by-point response letter to the Editor, the Associate Editor, and each individual referee, all in a single file.

Sincerely,

Po-Ling Loh

Editor, The Annals of Applied Statistics

A.2 Associate Editor Report

Both reviewers agree that this paper makes a nice methodological contribution, and that the motivation for the paper is compelling. However, this paper is not yet suitable for AoAS, given the applied and real world focus of the journal. The practical impact of the methodology is unclear, as the simulations do not clearly show this method is better, and the data analysis is not very thorough and it is unclear what the significance of this dataset is. Perhaps the authors should focus more on the data analysis and the downstream impact of the methods they propose.

This methodology is highly related to Camerlenghi et al (2022). This manuscript should more clearly articulate its original contributions and how they extend beyond existing work.

There are also multiple typos and confusing points as noted by the reviewers, that should be corrected.

A.3 Reviewer 1 Report

General comment. The authors introduce a novel Bayesian nonparametric framework for the prediction of count data. The paper is motivated by a clear and relevant application: predicting new users, total triggers, and the sufficient length of the follow-up period in the design of A/B tests (i.e., an online marketing methodology aimed at comparing the performance of two versions of a product or service). Building upon the literature on the Indian buffet process

and its generalizations, the authors’ approach yields closed-form posterior and predictive quantities that are both theoretically appealing and computationally efficient. In particular, parameter estimation and prediction under the proposed models do not require computationally expensive Markov chain Monte Carlo methods, as is often the case in Bayesian nonparametric models. This results in a theoretically elegant and practically flexible framework that can be applied to large datasets thanks to its computational efficiency. I thoroughly enjoyed reading this paper. It is well written and carefully organized, with strong applied motivation, solid methodological development, and convincing validation through extensive simulations and real-world experiments. However, I believe that some adjustments are needed to make the paper fully ready for publication; those are detailed below.

Point 1. While the paper is overall well written, I found Section 3.4 the most difficult to read and somewhat confusing. Most of the confusion likely stems from the fact that D_M , defined here as the number of follow-up days needed to see M new users, is set equal to $F'_{M-N_{D_0}}$, i.e., the day when the $(M - N_{D_0})$ -th new user triggered, which does not seem to align with the earlier definition. Is the term “new users” used interchangeably with “unobserved users” in this section? Do both terms refer to users not observed (i.e., not triggered) during the pilot study of length D_0 ?

Moreover, while the measure F' is clearly related to the measure Z^{FT} in Eq. (4), the presence of Theorem 3.8 in this section suggests that the reader would benefit from further clarification here either on the distributional assumptions for F' or through an explicit reference describing its relationship with Z^{FT} . Without such clarification, the statement of Theorem 3.8 appears somewhat vague.

Point 2. p.3, Eq. (1): the second sum should run from $D_0 + 1$ to $D_0 + D_1$, right?

Point 3. p.4, ll.8–10: the sentence starting from “this assumption clearly. . .” sounds too strong, or at least imprecise. At this point in the paper, no distributional assumption on the observations $A_{d,n}$ has been specified. Why is it obvious that all users out of the fixed total number M will eventually trigger as $D_1 \rightarrow \infty$?

Point 4. In Definition 3.1, the statement $A_{d,n} \mid \mu \stackrel{\text{iid}}{\sim} G(\cdot \mid \theta_n)$ would likely be clearer if written as $A_{d,n} \mid \mu \stackrel{\text{ind}}{\sim} G(\cdot \mid \theta_n)$, since the distribution on the right-hand side depends on n .

Point 5. On p.5, Eq. (5), there appears to be an s in place of a θ . Moreover, the subsequent phrasing “which puts to zero all values greater than D ” sounds too informal; consider rephrasing it more formally, for example, as “which assigns zero probability to all values exceeding D .”

Point 6. On p.5/6, I suggest mentioning that P_0 is a non-atomic distribution, either in the last paragraph of p.5 or within Definition 3.2.

Point 7. At the end of Definition 3.2, is $\tilde{\mu}$ supposed to be just μ ?

Point 8. In Theorem 3.4, was π_G previously defined?

Point 9. When specific theorems and the algorithm are referred to within the text, the words “theorem” and “algorithm” are sometimes capitalized and sometimes not.

Point 10. In Proposition 3.6, the sum defining $U_{D_0}^{(D_1)}$ should start from $D_0 + 1$.

Point 11. p.9: there is a typo and the word “users” is written as “usres.”

A.4 Reviewer 2 Report

The paper proposes an applied-methods framework to forecast user engagement in online A/B tests: (i) number of new/distinct users, (ii) total triggers including repeats, and (iii) days to reach target M users. It builds on stable-beta scaled process priors to model heavy-tailed user propensities and obtains closed-form posterior/predictive distributions that make the approach fast for large platforms.

Major concern (a). Although the context and the theoretical results are tailored to the specific application in A/B testing, the backbone of the proposed framework is essentially the same as in Camerlenghi et al (2022), that is, the scaled-process priors, in particular the stable-Beta scaled process (SB-SP), used to overcome the rigidity of standard CRM priors for “unseen features.” It’s clear that the overlap is mostly at the level of the prior construction and its posterior-predictive flexibility; the differences lie in inferential targets (unseen-feature estimation vs. multi-horizon forecasting and stopping rules) and the emphasis on deployable procedures and calibration for A/B experimentation workflows. This is my conclusion after reading both manuscripts; however, this manuscript should more clearly articulate its original contributions and how they extend beyond existing work.

Major concern (b). The manuscript is interesting, extensive, and technically careful, but my major concern is that its center of gravity is on probabilistic development: multiple likelihood types, several inferential targets, and a rich catalogue of distributional results. Therefore, it reads more like a methodological contribution adapted to an applied venue than a paper conceived with a truly applied motivation. The data analysis feels thin: a couple of datasets only superficially investigated, sparse diagnostics, and little evidence that the method has a practical impact and changes decisions in A/B testing (e.g., earlier stopping, fewer underpowered launches, calibrated risk at given service-level targets).

Major concern (c). The paper’s general objective is unclear. The introduction oscillates between (i) improving causal effect estimation and (ii) improving experimental sampling/design, but it never nails down a single primary goal,

the estimand, or the decision the method is meant to support. If the contribution is design, the paper must say what decision is being optimized and under which constraints, e.g., “choose D_0 (pilot length) and/or n_0 (pilot size) to minimize expected total time/cost to a pre-specified precision on the ATE.” If the true goal is causal estimation rather than design, the authors should define the estimand and demonstrate precision/coverage gains relative to standard estimators, treating design results as ancillary. Moreover, the data analysis should demonstrate the method’s impact and its practical advantage over competing methods in the context of A/B testing.

Major concern (d). Theoretical results are shown without comments about their consequences for the application. For example, after Corollary 3.5, it is mentioned that the predictive distribution of quantities that can be expressed as functionals of μ' will be the same under the Be-SSP and TG-SSP models. However, the practical consequences are not discussed. In Proposition 3.7, the Bayesian estimator for $T_{D_0}^{(D_1)}$ has a potentially interesting mixture form, e.g., in terms of rate ratios, but this is not commented upon.

Minor points and editorial issues.

(a) p.3: “One of the main challenges...crucial for accurately predicting user engagement.” The motivation for the paper would be clearer if those features were visualized in the data.

(b) p.3: “We can broadly categorize...linear programming approach.” It is not clear what the difference is between the proposed method and the approaches mentioned here, and why these approaches are limited.

(c) p.4: “Ionita-Laza et al. (2009); Richardson et al. (2022) assume that the total number of users, say M , is fixed and known.” Isn’t it always the case, though, that the number of users is known, in A/B testing? Isn’t there a cost to add days and subjects?

(d) p.4: “user activity often exhibits a power law growth.” Please, add a reference or a visualization.

(e) p.4: “Moreover, in some situations, focusing only on the first trigger times F_n might be preferable as we demonstrate in our experiments.” It looks like F_n can be derived from coarse data, so this point is not clear.

(f) The relevance of Definition 3.1 is clearly not related to the application. Moreover, it introduces concepts that may not be familiar to the reader, like the score distribution, which has not been previously defined.

(g) p.5: “Parametrizing the success probability of the negative binomial distribution...infinitesimally small values.” This sentence needs clarification.

(h) p.5: “the choice of the prior distribution for μ is crucial for capturing the complex dynamics and heterogeneity of user behavior.” This sentence needs clarification.

(i) Section 5.2 includes some comparisons with other methods. However, Figure 6 shows that across values of τ , the proposed Be-SSP has predictive

performance comparable to that of the Indian buffet process (IBP). Because the scaled-stable-beta-Bernoulli process (SSP) is not plotted, the claimed comparability to SSP cannot be evaluated. As a result, the empirical advantage of the proposed method is not clear in this setting.

(j) The real-data analysis would benefit from a fuller description of the dataset. Even if the data are proprietary, the paper should provide high-level information about the types of experiments conducted, the goals of the analysis, and why the proposed model is suitable for those aims. It is also unclear how the results would inform company decision-making, particularly relative to alternative modeling proposals.

(k) If it were for me, I would move most of the Simulations in the Supplementary Material, and focus the manuscript on the impact on the real data analysis.

(l) The authors correctly “highlight how the choice of the prior distribution for μ plays a crucial role for the quality of the posterior and predictive distributions of interest.” I would also add the choice of the score distribution. However, the manuscript does not thoroughly discuss how to assess model fit and select the optimal score distribution for a given problem.

Typos and inaccuracies. The paper contains numerous typos and inaccuracies. Some cross-references to equations and citations are unresolved (??). The paper needs an accurate reading. I am listing several instances here; however, the issues are present throughout the article.

- p.3: If $F_n = \min_d \mathbb{I}[A_{d,n} > 0]$ then the set of tuples does not need to include F_n . The data are simply represented by $A_{1:D,n}$.
- p.5, Section 3.2: “ $\Delta^{-\alpha} \sim \mathcal{E}(1)$ ”; $\mathcal{E}(1)$ is not defined.
- p.7, Corollary 3.5: What is $\tilde{\Delta}_{1,h}^{-\alpha}$ in (9)? It is not defined in the Corollary.
- p.11: “our strategy is more scalable as it does require performing Markov chain Monte Carlo simulations” $\rightarrow$ “does *not* require.”
- p.16: “Moreover, the length of the intervals increases with parameter tau of the data generating process” $\rightarrow$ the parameter τ .